

Arash Rocky

Department of Electrical & Computer Engineering

University of Windsor, 401 Sunset Ave.

Windsor, ON N9B 3P4

T: +1 226-758-9767 | rocky@uwindsor.ca | arashrocky.github.io/ | Google Scholar

Objective

Dedicated and highly skilled PhD Candidate specializing in Computer Vision for Autonomous Vehicles and Advanced Driving Assistance Systems (ADAS). Seeking to leverage research experience and technical expertise in a challenging role within an innovative organization.

Education

PhD Candidate in Electrical and Computer Engineering

University of Windsor, Canada

2021 - Present

MASc in Electrical Engineering, Electronics

Shahid Chamran University of Ahvaz

2010 - 2013

Thesis: *Image Enhancement Using Super-Resolution* (Outstanding Final Thesis)

BSc in Electrical Engineering, Electronics

Shahid Chamran University of Ahvaz

2004 - 2009

Thesis: *Evaluation of EEG signals using Artificial Neural Networks* (Outstanding Thesis)

Skills

- **Programming:** Python, C++
- **Tools/Software:** PyTorch, Keras, TensorFlow, OpenCV, NumPy, Pandas, Scikit-learn, CUDA, MATLAB
- **Research Areas:** Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Image Processing, Multiple Object Tracking, Video Instance Segmentation, Vision-Language Model (VLM), Accident Detection and Anticipation, EEG Signal Processing

Research Experience

Research Assistant

September 2021 - Present

Department of Electrical and Computer Engineering, University of Windsor, Canada

- Pursuing research on Traffic Accident Detection Using Dashcam Videos with deep learning approaches.
- Utilizing object detection, object tracking, instance segmentation and VLMs for accident detection.

Graduate Assistant

September 2021 - May 2025

Department of Electrical and Computer Engineering, University of Windsor, Canada

- Conducted laboratory sessions and tutorials.
- Prepared teaching materials, assignments, lectures, and marked assignments, projects, and exams.
- Courses: Image Processing, Digital Logic Design, Capstone Projects, Computer System Architecture

Teaching Experience

Lecturer

July 2012 - June 2013

Department of Electrical and Computer Engineering, Karoon Institute of Higher Education, Ahvaz, Iran

- Courses: Electrical Circuits I, Electronics I, Logic Circuits

Guest Lecturer

2011

Department of Electrical Engineering, Shahid Chamran University, Ahvaz, Iran

- Topic: What Is Super-Resolution?

Guest Lecturer

2010

Department of Electrical Engineering, Shahid Chamran University, Ahvaz, Iran

- Topic: Introduction to 4th Generation Mobile Networks

Research Topics

- **Traffic Accident Detection Using Dashcam Videos:** Developing new baseline for accident detection using object detection, tracking, instance segmentation, and depth map (Published one journal).
- **Emotion Assessment Using EEG Signal Classification:** Using synthetic statistical-frequency feature extraction and feature selection for neural network signal classification (Published one journal).
- **Image Enhancement Using Super-Resolution:** Implemented a super-resolution model, Generalized Non-Local Means, by fusion of denoising, deblurring, and upscaling (Published one paper).

Publications

1. **Rocky A.**, Wu Q.M.J., "Towards Automatic Video Annotation with ASH: Zero-Shot Open-Vocabulary Multi-Object Tracking and Segmentation", *arXiv preprint arXiv:2610.01022*, 2026
2. **Rocky A.**, Wu Q.M.J., "SAM2Auto: Auto Annotation Using FLASH", *arXiv preprint arXiv:2506.07850*, 2025
3. **Rocky A.**, Wu Q.M.J., Zhang W., "Review of Accident Detection Methods Using Dashcam Videos for Autonomous Driving Vehicles", *IEEE Transactions on Intelligent Transportation Systems*, 2024
4. Saba V., **Rocky A.**, "A real-time electroencephalography classification in emotion assessment based on synthetic statistical-frequency feature extraction and feature selection", *Annals of Military and Health Sciences Research*, 2016
5. **Rocky A.**, Ansari-Asl K., Akbarizadeh G., "The study on Super-Resolution Algorithm with Implicit Motion Estimation", *Iranian Conference on Intelligent Systems*, 2013
6. Amin R., Marashi SMH., Noori SMR., et al., "Medical, pharmaceutical, and nutritional applications of 3D-printing technology in diabetes", *Elsevier*, 2024

Honors

- Research Assistantship Award, University of Windsor, Canada, 2021 - Present

Professional Services

- Reviewer for IEEE Transactions on Cybernetics, 2019 - Present
- Reviewer for IEEE Transactions on Circuits and Systems for Video Technology, 2018 - Present